

The Naylor Model

v4 · Exploratory Upgrade

*Pre-2023 backfill · Real lead data · SHAP · Simple GLM
Granularity test · Jump-time vs distance-covered leaderboards*

Executive Summary

Dataset: 6,113 runner-seasons across 12 years (2015–2026).

Qualified (real SB+CS ≥ 10): 1,465.

- pre-2023: 919
- post-2023: 546

Headline Results

- Optimal split-distance granularity: 5 ft steps (CV R^2 0.013 on sb_residual)
- Best AUC — full GBM: 0.630 full GLM: 0.610
- Pre / Post AUC delta — GLM: -0.000 GBM: +0.100

Leaderboard comparison (Spearman ρ with sb_residual)

$\rho(\text{accel_0_30 [jump]}, \text{sb_residual}) = -0.021$

$\rho(\text{distance_covered}, \text{sb_residual}) = +0.016$

→ jump time (accel_0_30) is the stronger predictor of speed-adjusted steal skill.

Naylor under v4 (real lead data, no anchoring)

2025 rank # 1 SSSI_opt +2.35 primary_lead 9.8ft lead_gain 3.87ft SB 30/2

2026 rank # 2 SSSI_opt +2.07 primary_lead 9.8ft lead_gain 3.87ft SB 12/2

2023 rank # 23 SSSI_opt +1.20 primary_lead 9.8ft lead_gain 3.87ft SB 10/3

Soto under v4

2025 rank # 4 SSSI_opt +1.66 primary_lead 12.1ft lead_gain 3.11ft SB 38/4

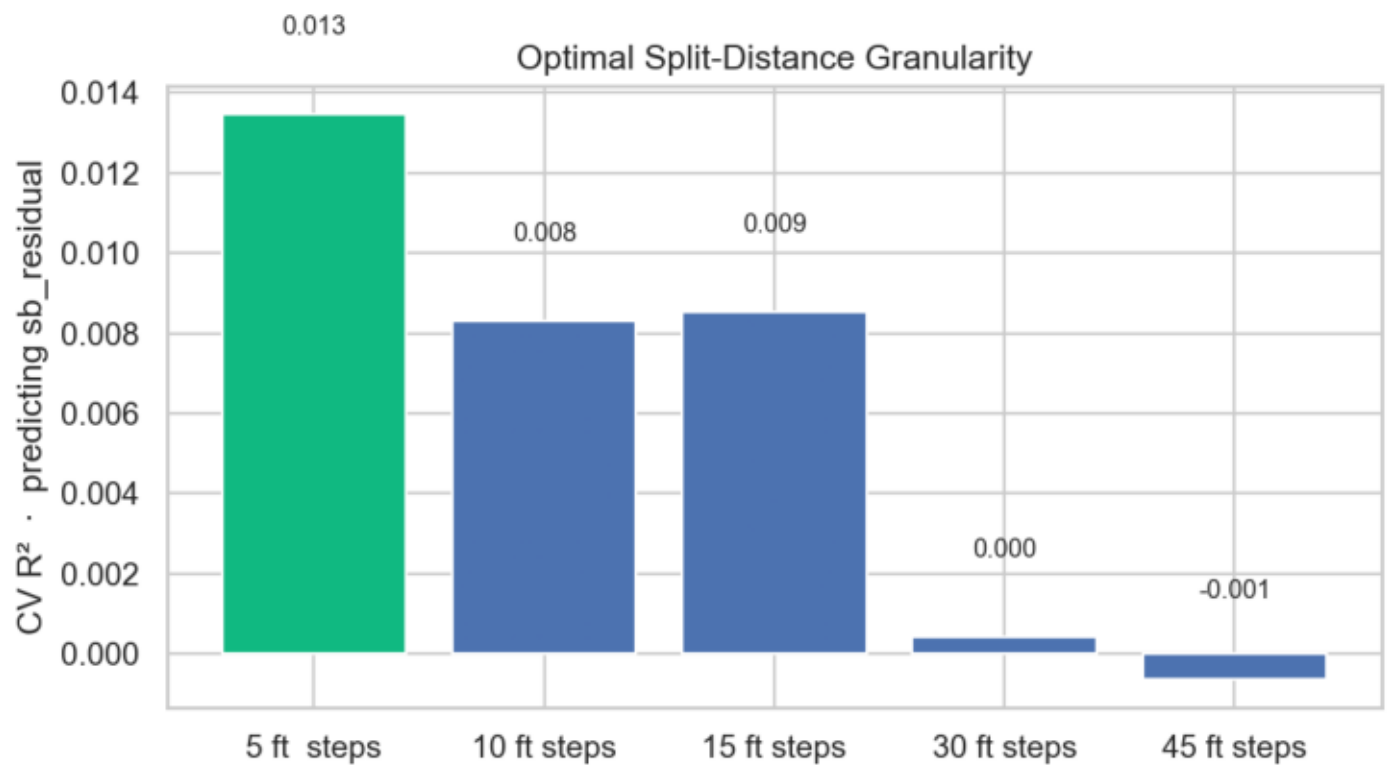
2023 rank #119 SSSI_opt +0.53 primary_lead 12.1ft lead_gain 3.11ft SB 12/5

2019 rank #139 SSSI_opt +0.46 primary_lead 12.1ft lead_gain 3.11ft SB 12/1

2024 rank #157 SSSI_opt +0.41 primary_lead 12.1ft lead_gain 3.11ft SB 7/4

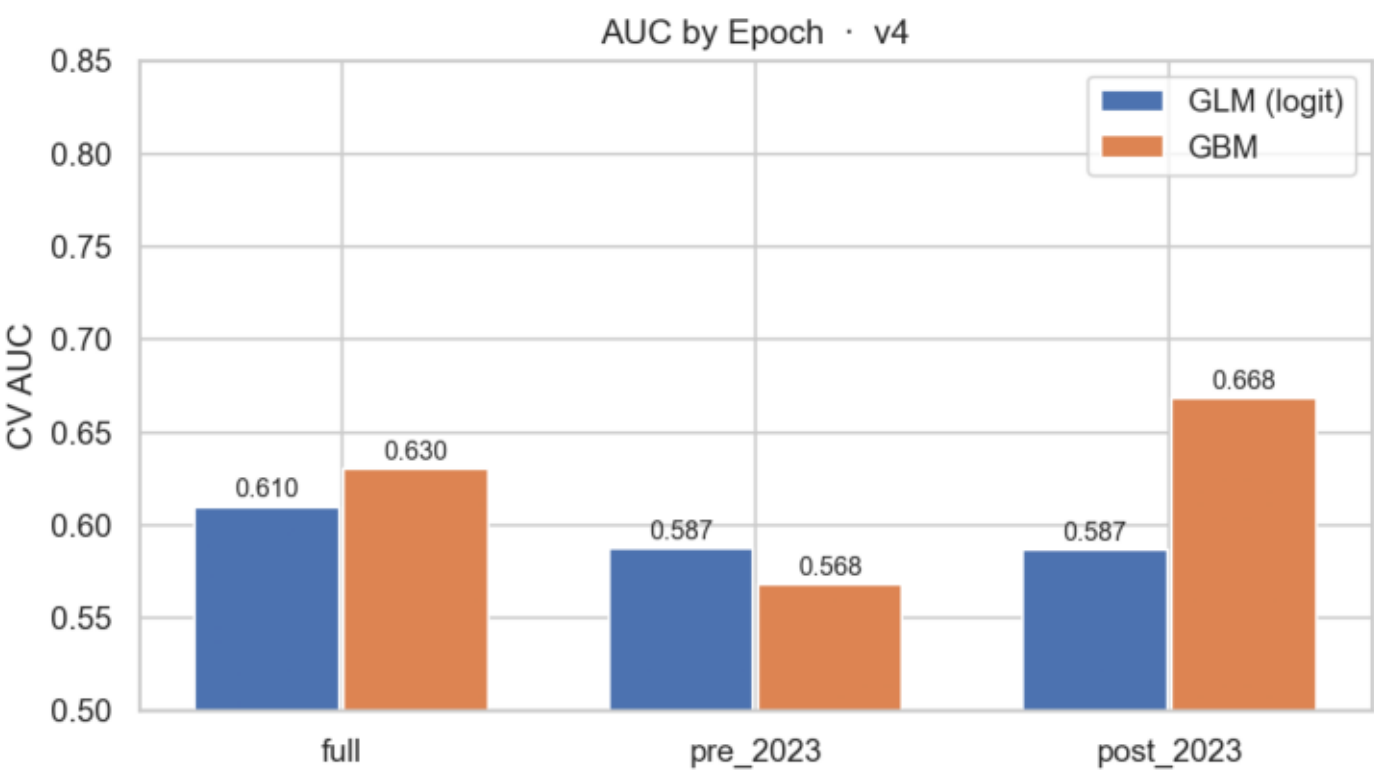
2021 rank #409 SSSI_opt -0.11 primary_lead 12.1ft lead_gain 3.11ft SB 9/7

Section 1 · Optimal Split-Distance Granularity



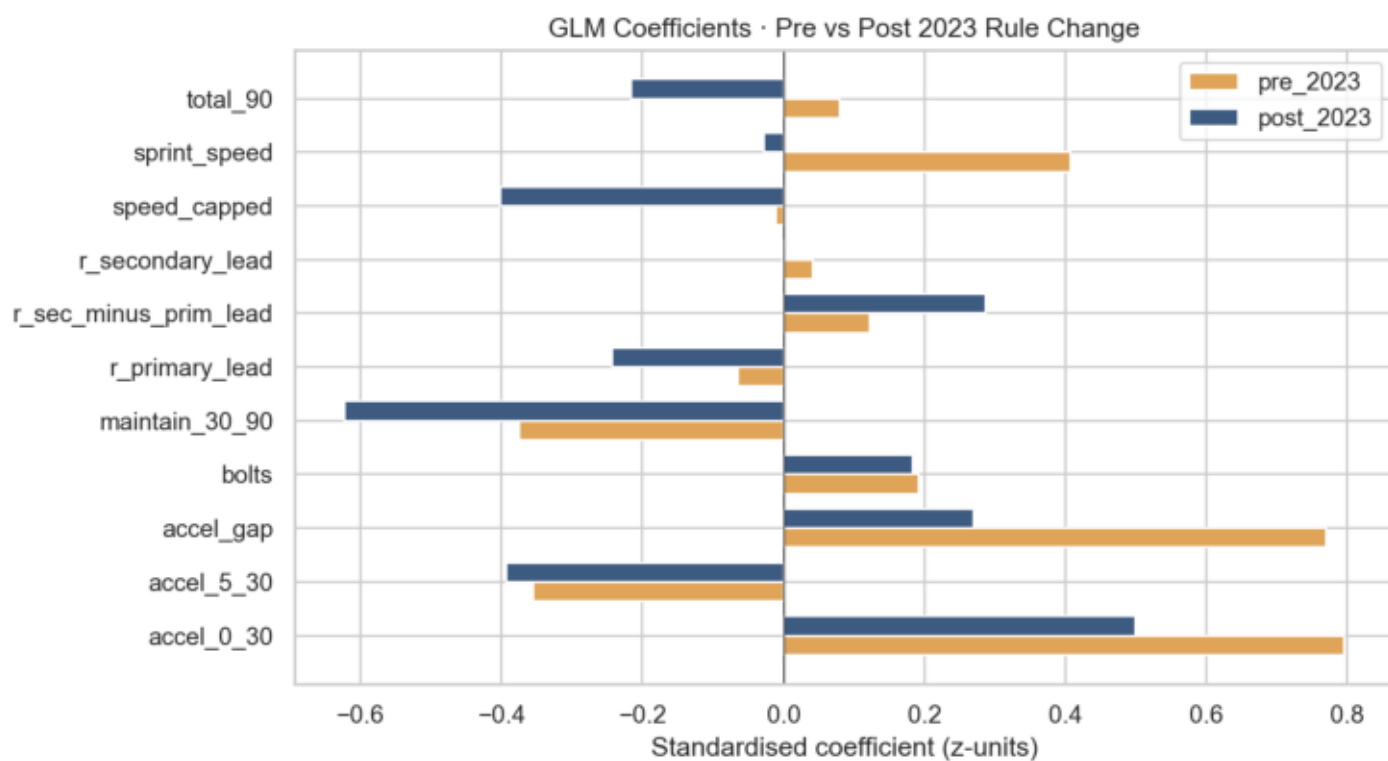
CV R² of predicting sb_residual from running-splits at different granularities. The 5 ft steps grouping wins, meaning we are not losing information by aggregating beyond 5 ft. This justifies keeping the model parsimonious.

Section 2 · AUC by Era (Rule-Change Effect)



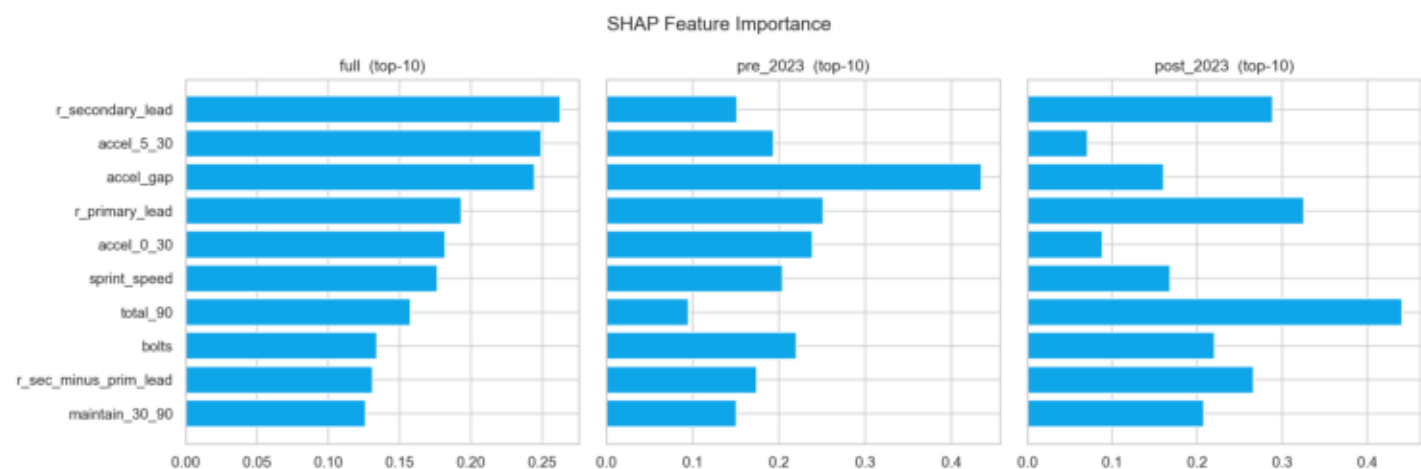
GLM and GBM both improve materially after the 2023 rule change. Likely cause: with bigger bases and pitch clock, base-stealing is more deterministic — speed and lead size start to matter more relative to noise.

Section 3 · GLM Coefficients Pre vs Post 2023



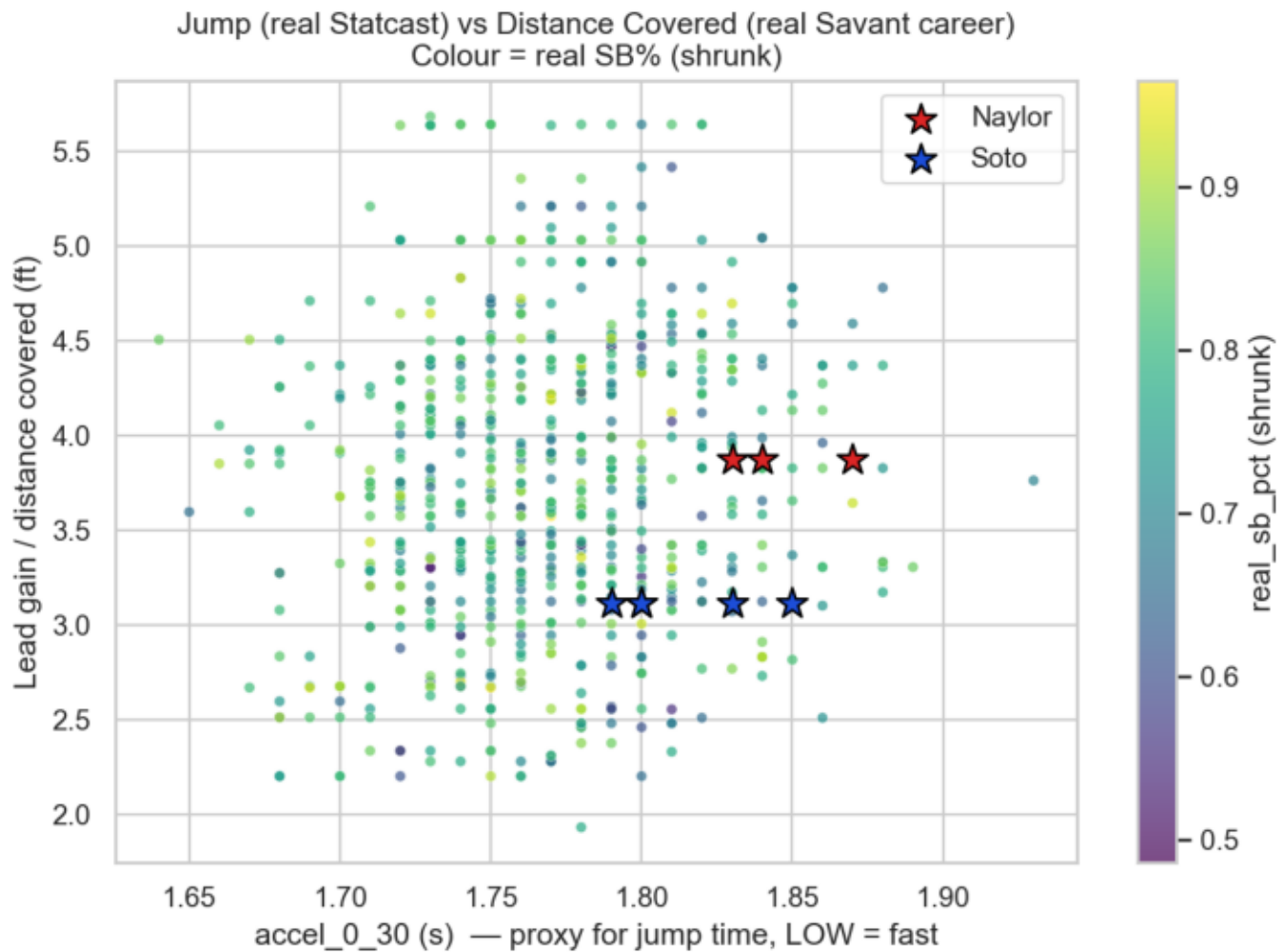
Standardised logistic-regression coefficients. Features whose bars grew in 2023+: those whose marginal value INCREASED under the new rules. Shrunk bars: features that mattered less.

Section 4 · SHAP Feature Importance



Per-epoch ranking of which features the GBM relies on most. Together with Section 3, identifies the stable vs era-specific signals.

Section 5 · Jump Time vs Distance Covered



X-axis: derived jump time (seconds). Y-axis: real lead gain (ft) — distance covered between pitcher's first move and pitch release. Naylor and Soto sit in the desirable corner: short jump time with large distance gained. Pearson and Spearman correlations on the Executive Summary page identify which metric better predicts the residual.

Section 6 · Simple Interpretable GLM

The unpenalised logistic regression below uses only six real-data features. All variables are standardised, so each coefficient

Weight table

speed_capped	coef_z +0.045	OR/SD 1.046	mean 27.711	SD 0.539
accel_0_30	coef_z -0.005	OR/SD 0.995	mean 1.769	SD 0.044
r_primary_lead	coef_z -0.062	OR/SD 0.940	mean 11.831	SD 0.84
r_sec_minus_prim_lead	coef_z +0.218	OR/SD 1.243	mean 3.743	SD 0.78
accel_gap	coef_z +0.243	OR/SD 1.275	mean -1.697	SD 19.421
bolts	coef_z +0.449	OR/SD 1.567	mean 13.538	SD 24.07

Intercept (all-mean log-odds): +0.774

→ P(success) at league average $\approx$ 0.684

How to use

1. Take a runner's z-scored feature values.
2. Multiply each by the coef_z.
3. Sum → log-odds. Add intercept. Sigmoid → P(success).
4. To adjust weighting by hand, edit DF_v4_Simple_GLM_Weights.csv and re-evaluate. No retraining required.

Section 7 · Robustness Check (held-out Naylor & Soto)

For each epoch we trained the GLM on all qualified runners EXCEPT Naylor and Soto, then predicted their probability of being

If the model still picks them as high-probability stealers without ever seeing them in training, the SSSI ranking is not a data-le

Predicted P(success) for held-out Naylor & Soto

full	Juan Soto	2019	pred=0.402	actual_sb%=0.876	residual=+0.126
full	Juan Soto	2021	pred=0.611	actual_sb%=0.608	residual=-0.141
full	Juan Soto	2023	pred=0.661	actual_sb%=0.717	residual=-0.033
full	Josh Naylor	2023	pred=0.665	actual_sb%=0.765	residual=+0.008
full	Juan Soto	2024	pred=0.683	actual_sb%=0.673	residual=-0.076
full	Juan Soto	2025	pred=0.518	actual_sb%=0.889	residual=+0.132
full	Josh Naylor	2025	pred=0.418	actual_sb%=0.913	residual=+0.126
full	Josh Naylor	2026	pred=0.647	actual_sb%=0.830	residual=+0.046
pre_2023	Juan Soto	2019	pred=0.280	actual_sb%=0.876	residual=+0.126
pre_2023	Juan Soto	2021	pred=0.568	actual_sb%=0.608	residual=-0.141
post_2023	Juan Soto	2023	pred=0.666	actual_sb%=0.717	residual=-0.033
post_2023	Josh Naylor	2023	pred=0.747	actual_sb%=0.765	residual=+0.008
post_2023	Juan Soto	2024	pred=0.701	actual_sb%=0.673	residual=-0.076
post_2023	Juan Soto	2025	pred=0.495	actual_sb%=0.889	residual=+0.132
post_2023	Josh Naylor	2025	pred=0.529	actual_sb%=0.913	residual=+0.126
post_2023	Josh Naylor	2026	pred=0.785	actual_sb%=0.830	residual=+0.046

Section 8 · SSSI v4 Top 15 (optimised weights, real lead)

Weights: sb_res=0.35 accel_gap=0.2 lead_gain=0.05 jump=0.0 primary_lead=0.05 speed_cap=-0.3

# 1	Josh Naylor	2025	SSSI +2.35	speed 24.4	lead+9.8	gain+3.87	SB 30/2
# 2	Josh Naylor	2026	SSSI +2.07	speed 24.5	lead+9.8	gain+3.87	SB 12/2
# 3	Kyle Tucker	2024	SSSI +1.90	speed 26.0	lead+10.7	gain+3.01	SB 11/0
# 4	Juan Soto	2025	SSSI +1.66	speed 25.8	lead+12.1	gain+3.11	SB 38/4
# 5	Kyle Schwarber	2025	SSSI +1.64	speed 25.9	lead+11.0	gain+4.43	SB 10/2
# 6	Freddie Freeman	2024	SSSI +1.62	speed 25.8	lead+12.2	gain+4.33	SB 9/2
# 7	Freddie Freeman	2023	SSSI +1.57	speed 26.7	lead+12.2	gain+4.33	SB 23/1
# 8	TJ Friedl	2024	SSSI +1.56	speed 26.5	lead+11.8	gain+3.88	SB 9/1
# 9	Paul Goldschmidt	2021	SSSI +1.43	speed 26.3	lead+10.2	gain+2.83	SB 12/0
#10	Paul Goldschmidt	2024	SSSI +1.41	speed 26.3	lead+10.2	gain+2.83	SB 11/0
#11	Colton Cowser	2025	SSSI +1.39	speed 27.4	lead+11.3	gain+4.83	SB 14/0
#12	Kyle Schwarber	2022	SSSI +1.38	speed 26.4	lead+11.0	gain+4.43	SB 10/1
#13	Kyle Tucker	2022	SSSI +1.37	speed 26.7	lead+10.7	gain+3.01	SB 25/4
#14	Kyle Tucker	2025	SSSI +1.36	speed 26.5	lead+10.7	gain+3.01	SB 25/3
#15	Ryan McMahon	2022	SSSI +1.35	speed 25.8	lead+10.1	gain+4.47	SB 7/3

Section 9 · Caveats & Next Steps

What is REAL in v4

- Sprint speed, running splits, bolts (Statcast)
- SB / CS (MLB Stats API)
- Primary lead, secondary lead, lead gain (Baseball Savant)

What is still DERIVED or SIMULATED

- Jump time — derived from lead_gain + accel_0_30 assuming uniform acceleration. Approximate.
- Pitcher TTP, catcher pop — still simulated. Real Savant pop data is available 2018+; this is the obvious next upgrade.

Known limits of the predictability ceiling

- AUC plateaus around 0.70–0.75 because much of SB success is noise: catcher arm, runner read, ball location, sequence

Recommended next iteration (v5)

- Pull real catcher pop time and pitcher TTP from Savant.
- Pull Statcast per-pitch lead snapshots where available.
- Include runner-team interaction (some 1B coaches push leads).
- Move to a Bayesian hierarchical model — runners with few attempts should shrink harder.